

the maximum index  $n$  across all multisets is 2. to achieve this, the smallest possible maximum element  $m$  is 4. among all multisets with index 2 and maximum element 4, the smallest number of elements is 4. the multiset  $\{1, 2, 3, 4\}$  satisfies the conditions for index 2 and has four elements. for  $i = 1$ , partition into  $b_1 = \{1, 3\}$  and  $a_1 = \{2, 4\}$ , since  $a_1 = 1 + b_1$ . for  $i = 2$ , partition into  $b_2 = \{1, 2\}$  and  $a_2 = \{3, 4\}$ , since  $a_2 = 2 + b_2$ . no smaller multiset achieves index 2. thus the smallest number of elements is 4.

$\boxed{4}$